

Cardiovascular Disease Prediction

Exploratory Data Analysis, Data Cleaning, and Logistic Regression

Coursework project team

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Abstract

This project investigates whether the presence of cardiovascular disease can be predicted from demographic, anthropometric, clinical, and lifestyle variables. The analysis uses the Kaggle *Cardiovascular Disease Dataset*, containing approximately 70,000 records and 13 original attributes. It follows a reproducible workflow consisting of exploratory data analysis (EDA), data cleaning, feature engineering, and logistic-regression baseline modelling. The main objective is to examine how implausible measurements and outliers affect statistical relationships and model performance. After conservative physiologically motivated cleaning and the addition of body-mass index (BMI) and pulse pressure, the reported ROC-AUC increases from approximately 0.76 to approximately 0.80. The project is educational only: it is not clinically validated and must not be used for medical decision-making.

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1 Introduction

This project examines how accurately the presence of cardiovascular disease, represented by the target variable `cardio`, can be predicted from patient-related features. The data source is the *Cardiovascular Disease Dataset*, which contains roughly 70,000 observations and 13 original attributes [1]. Its structure is typical of medical screening data: demographic variables such as age and sex, body measurements such as height and weight, clinical measurements such as blood pressure, cholesterol and glucose, and lifestyle indicators for smoking, alcohol use, and physical activity.

The aim is to construct and assess a transparent pipeline for EDA, cleaning, and baseline modelling. The project has four connected goals:

1. Explore the dataset to reveal its quality, distributions, and risks such as outliers and invalid encodings.
2. Clean the data by removing or correcting physiologically implausible values, including invalid blood-pressure entries and extreme body measurements.
3. Train an interpretable logistic-regression baseline.
4. Compare the raw and cleaned datasets in order to assess how data quality affects model performance and reliability.

The analysis was carried out in Jupyter notebooks. A code repository should contain the notebooks, the generated figures, environment requirements, and instructions to obtain the data. It should not redistribute the raw CSV file unless its licence and provenance have been independently confirmed to permit that use.

2 Foundations

2.1 Cardiovascular diseases

Cardiovascular diseases (CVDs) are diseases of the heart and blood vessels. They include coronary heart disease, myocardial infarction, stroke, and complications of arterial hypertension. CVDs have been among the leading causes of death worldwide for many years [2]. As a result, medicine has long sought not only to treat these conditions but also to estimate risk early enough to support prevention.

Early risk assessment relied mainly on known factors such as age and lifestyle. These relationships were later studied systematically in large population studies, including the Framingham Study [3]. Such evidence contributed to standardised risk models, for example SCORE, which estimate a person's cardiovascular risk with statistical methods [4].

In this project, cardiovascular disease is represented in a deliberately simplified form. The dataset supports a binary classification:

`cardio` = 0 indicates no diagnosed cardiovascular disease,
`cardio` = 1 indicates the presence of cardiovascular disease.

This abstraction allows classical classification methods, especially logistic regression, but it does not distinguish the disease type, severity, or stage.

2.2 Ethical context

The dataset is used primarily for teaching and learning in data science and machine learning. The purpose of this work is to learn and apply analytical methods in a realistic example, not

to derive clinically valid or medically actionable conclusions. The findings must therefore be interpreted only in a methodological context.

Misclassification has serious implications in real healthcare. In particular, a false negative—a person with disease classified as healthy—could delay further diagnosis or treatment. The developed model is consequently an educational illustration of EDA, data cleaning, and modelling. It neither replaces clinical expertise nor claims medical validity.

The sample is also not fully representative. It contains a substantial overrepresentation of records coded as female, which can affect model estimates and limit generalisation to other populations.

2.3 Privacy

The project uses a publicly available secondary dataset described as anonymised. It contains no direct identifiers such as names, addresses, or exact dates of birth. The analysis therefore does not seek to re-identify any individual. Nonetheless, anonymisation alone does not resolve the question of data-reuse rights; the source data should be linked to rather than republished unless its licence explicitly permits redistribution.

3 Dataset Description and Exploratory Data Analysis

3.1 Data source

The source is the Kaggle *Cardiovascular Disease Dataset* [1]. Kaggle is an open platform for data-science competitions and for sharing datasets used for learning and practice. With roughly 70,000 observations, the dataset is large enough for exploratory analysis and for training and evaluating basic machine-learning models.

The data contains no missing values, allowing all variables to be analysed consistently. At the same time, the EDA reveals numerous outliers and implausible values, making systematic cleaning necessary. Documentation about the original data collection, the exact study population, and data generation is limited. This limitation means that the project uses the dataset only for didactic purposes and does not make clinically generalisable claims.

3.2 Format and variables

The data is stored in CSV format and read from `cardio_train.csv`. In its raw form, it has 70,000 observations and 13 columns. The target variable `cardio` is binary and represents the presence (1) or absence (0) of cardiovascular disease.

Table 1: Feature groups in the raw dataset.

Group		Variables and interpretation
Continuous	measurements	<code>height</code> , <code>weight</code> , <code>ap_hi</code> , and <code>ap_lo</code> .
Ordinal	categories	<code>cholesterol</code> and <code>gluc</code> , each coded from 1 to 3.
Binary	indicators	<code>smoke</code> , <code>alco</code> , <code>active</code> , and <code>cardio</code> .
Categorical	code	<code>gender</code> $\in \{1, 2\}$.

All variables are numerically encoded, so the dataset is formally straightforward to model. However, numerical encoding does not ensure semantic correctness. The binary target is also a simplified representation of clinical reality: it supports a disease-present versus disease-absent classification but provides no information about diagnosis, severity, or progression.

3.3 Feature overview

The variables can be grouped into demographic, clinical, and lifestyle features. The demographic features include age, which is stored in days in the raw data, and the categorical **gender** code (1 = female, 2 = male). Height is measured in centimetres and weight in kilograms. For clearer interpretation and comparison, age is later transformed from days to years.

Clinical features include systolic blood pressure (**ap_hi**), diastolic blood pressure (**ap_lo**), cholesterol (**cholesterol**), and glucose (**gluc**). Blood pressure is represented as a numerical value in mmHg, whereas cholesterol and glucose are ordinal categories with discrete medical levels.

Lifestyle features describe smoking (**smoke**), alcohol use (**alco**), and physical activity (**active**). These are binary, self-reported indicators. The combination of the three feature groups provides a structured basis for subsequent EDA, cleaning, and modelling.

Feature Name	Category	Column Name	Data Type / Values	Description
Age	Objective Feature	<code>age</code>	int (days)	Patient age (in days)
Height	Objective Feature	<code>height</code>	int (cm)	Patient height (centimeters)
Weight	Objective Feature	<code>weight</code>	float (kg)	Patient weight (kilograms)
Gender	Objective Feature	<code>gender</code>	categorical code	1: woman, 2: man
Systolic blood pressure	Examination Feature	<code>ap_hi</code>	int	Systolic blood pressure (mm Hg)
Diastolic blood pressure	Examination Feature	<code>ap_lo</code>	int	Diastolic blood pressure (mm Hg)
Cholesterol	Examination Feature	<code>cholesterol</code>	1: normal, 2: above normal, 3: well above normal	Cholesterol level
Glucose	Examination Feature	<code>gluc</code>	1: normal, 2: above normal, 3: well above normal	Glucose level
Smoking	Subjective Feature	<code>smoke</code>	binary (0: no, 1: yes)	Whether patient smokes or not
Alcohol intake	Subjective Feature	<code>alco</code>	binary (0: no, 1: yes)	Whether patient consumes alcohol
Physical activity	Subjective Feature	<code>active</code>	binary (0: no, 1: yes)	Whether patient is physically active
Cardiovascular disease (target)	Target Variable	<code>cardio</code>	binary (0: no, 1: yes)	Presence (1) or absence (0) of disease

Figure 1: Overview of the features used in the analysis.

3.4 Univariate analyses

3.4.1 Demographic and anthropometric variables

The age distribution is shifted toward older participants. Most records fall between 40 and 64 years, with a peak around 55–60 years. There are substantially fewer observations below age 40, which is plausible for a cardiovascular-focused dataset.

Weight is approximately normally distributed but has a right-hand tail, indicating positive skewness. Most observations fall roughly between 55 and 110 kg. Individual very low and very high values are conspicuous and should be checked for plausibility.

Height values are also approximately normally distributed. Most observations lie between 150 and 180 cm, with a peak around 165–170 cm. A small number of values below 125 cm or above 200 cm may be outliers.

The sex distribution is not balanced: approximately 65% of records are coded as female and 35% as male. This imbalance should be considered during both modelling and interpretation

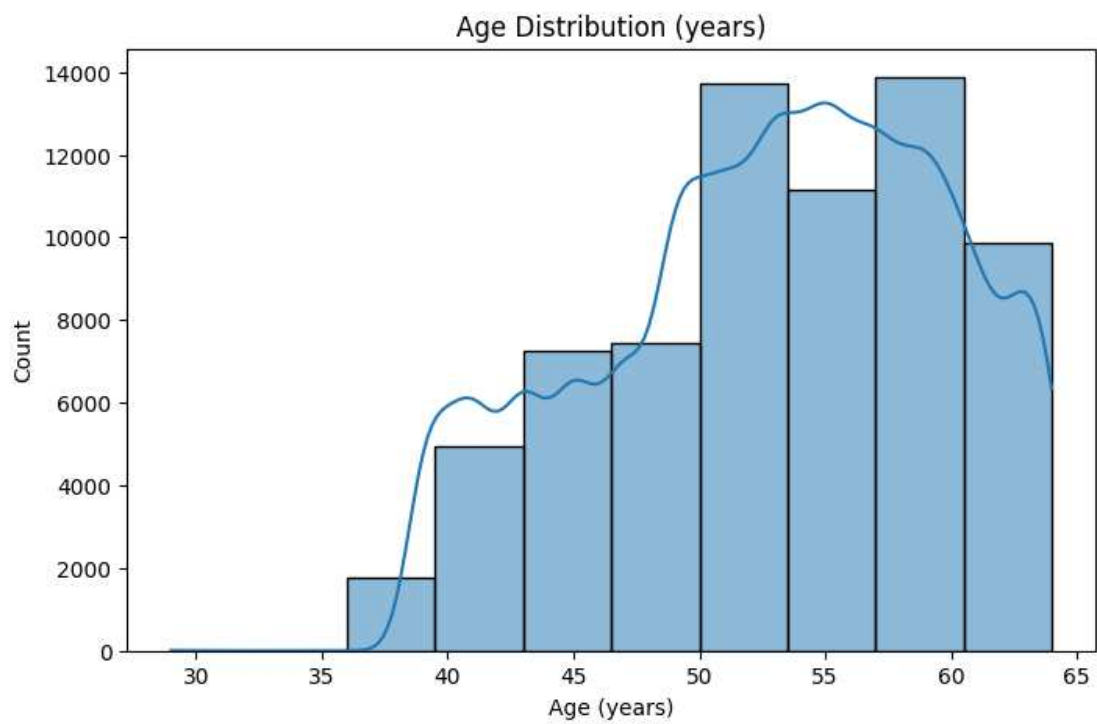


Figure 2: Age distribution in years.

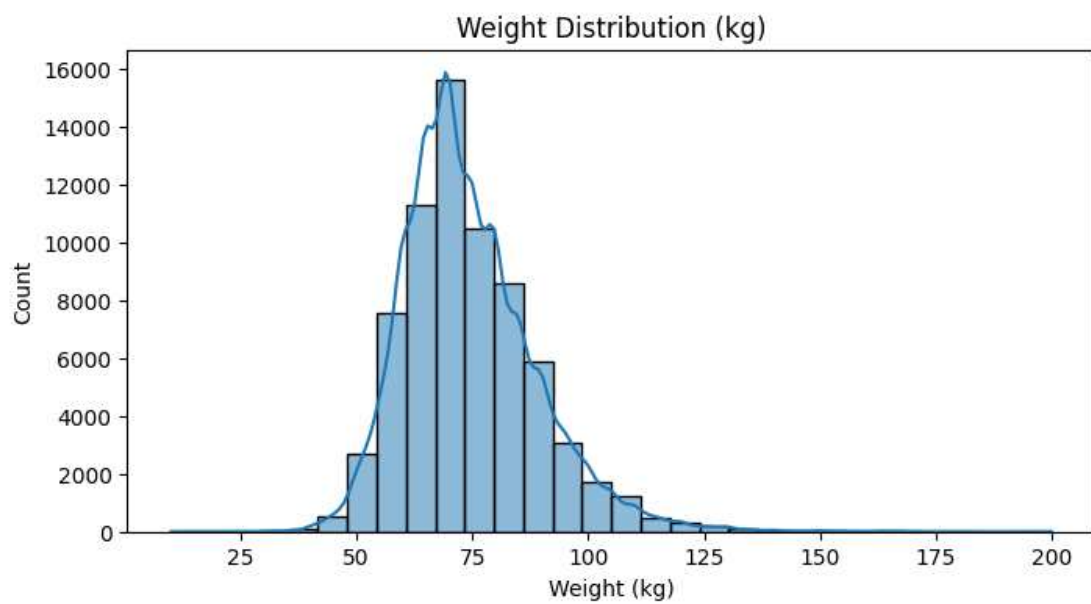


Figure 3: Weight distribution in kilograms.

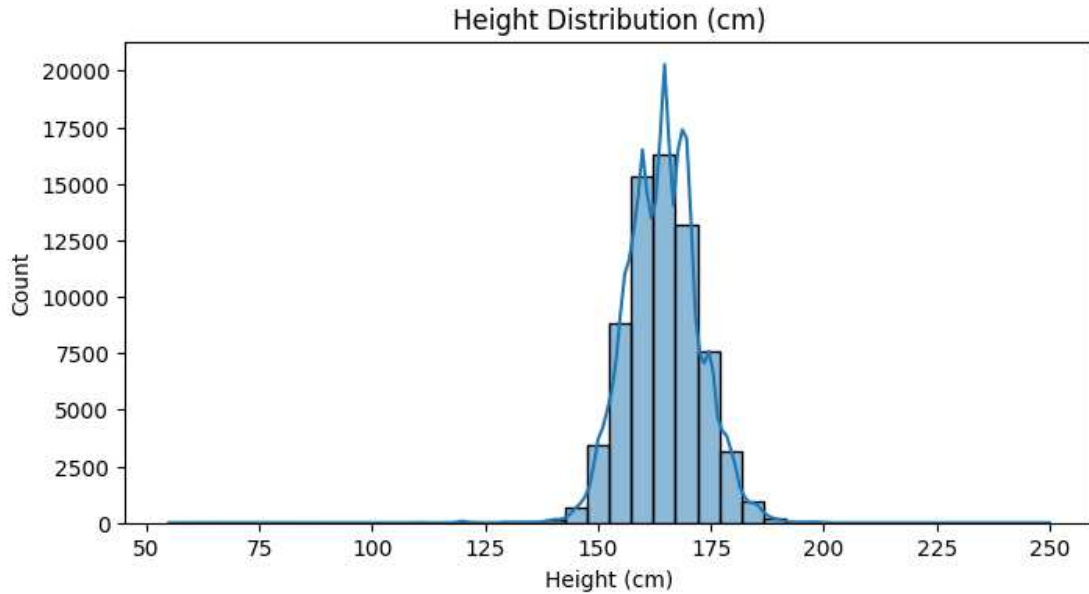


Figure 4: Height distribution in centimetres.

because it can reduce the transferability of results.

The target classes are nearly balanced. Approximately half of the records have `cardio=0` and half have `cardio=1`, so a strong class-imbalance effect is not expected.

3.4.2 Clinical variables

Most observations are in cholesterol category 1 (normal). Categories 2 (above normal) and 3 (well above normal) occur much less frequently. The same broad pattern applies to glucose: most observations are at category 1, while categories 2 and 3 are relatively small groups, though somewhat more common than their cholesterol counterparts.

For systolic blood pressure, the largest groups are in the 120–139 mmHg and 140–159 mmHg ranges. Low values (80–119 mmHg) and very high values (160 mmHg and above, especially 200 mmHg and above) are less common. Diastolic pressure is concentrated mainly between 80 and 99 mmHg; very low values below 80 mmHg and high values at 100 mmHg or above are rarer. The presence of extreme values in both variables is nevertheless visible and requires further investigation.

3.4.3 Lifestyle variables

The lifestyle variables are binary. Approximately 91.2% of records indicate no smoking and 8.8% indicate smoking. Approximately 94.6% indicate no alcohol use and 5.4% indicate alcohol use. Physical activity is less imbalanced: around 80.4% report being active and 19.6% report not being active. The low prevalence of smoking and alcohol use can limit how much isolated information these variables provide to a model.

3.5 Bivariate analyses with the target variable

The bivariate analysis examines selected features in relation to `cardio`. Its purpose is descriptive: each feature is considered separately to explore whether and how the proportion of records labelled with cardiovascular disease changes. It does not establish causal effects.

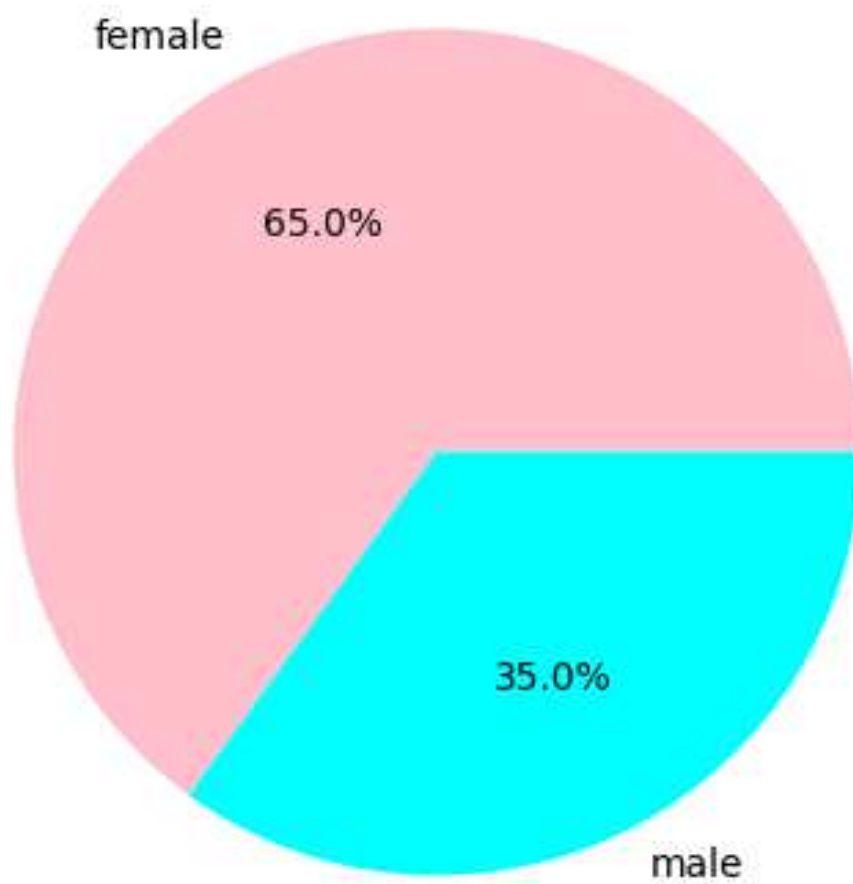


Figure 5: Distribution of the gender code in the dataset.

Proportion of Patients with/without Cardiovascular Disease

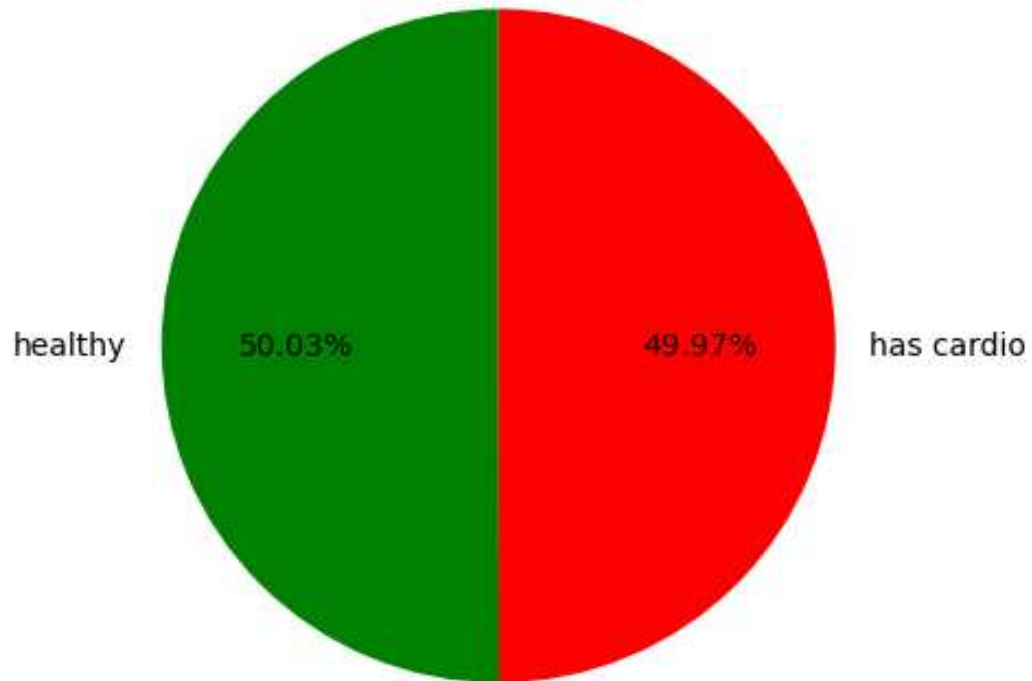
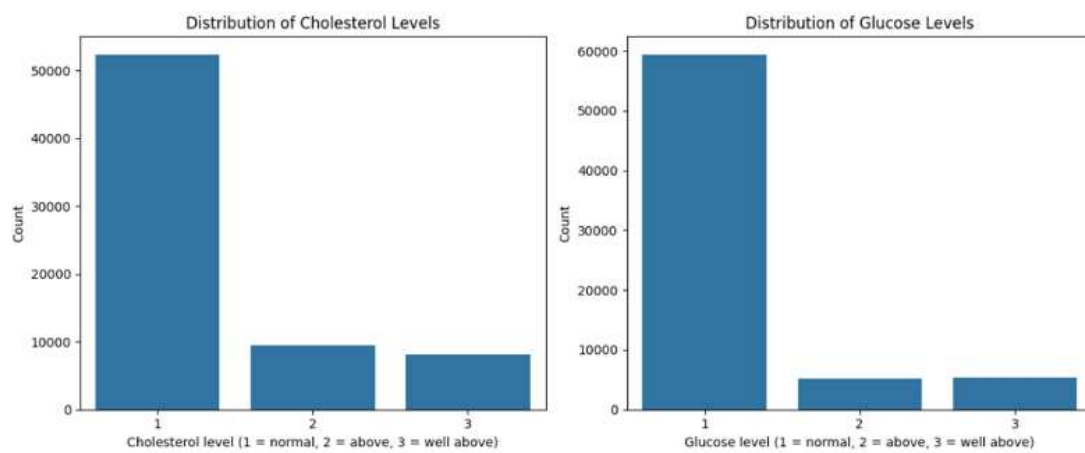
Figure 6: Class distribution of the target variable `cardio`.

Figure 7: Distribution of cholesterol and glucose categories (1–3).

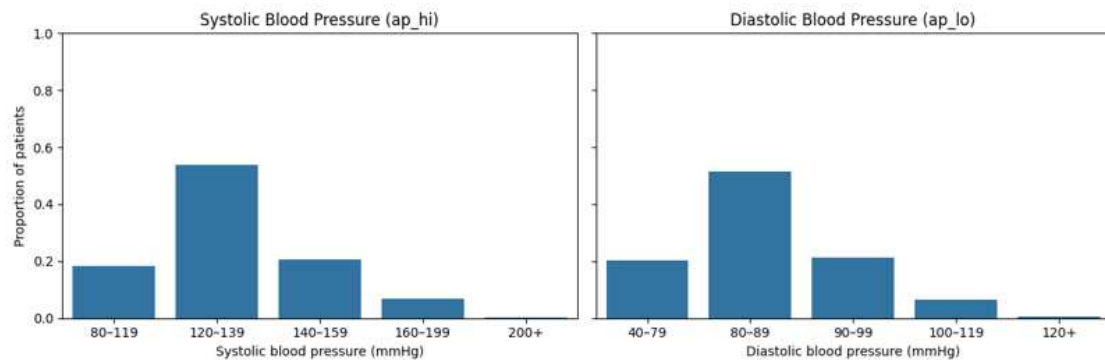


Figure 8: Distribution of systolic (`ap_hi`) and diastolic (`ap_lo`) blood pressure.

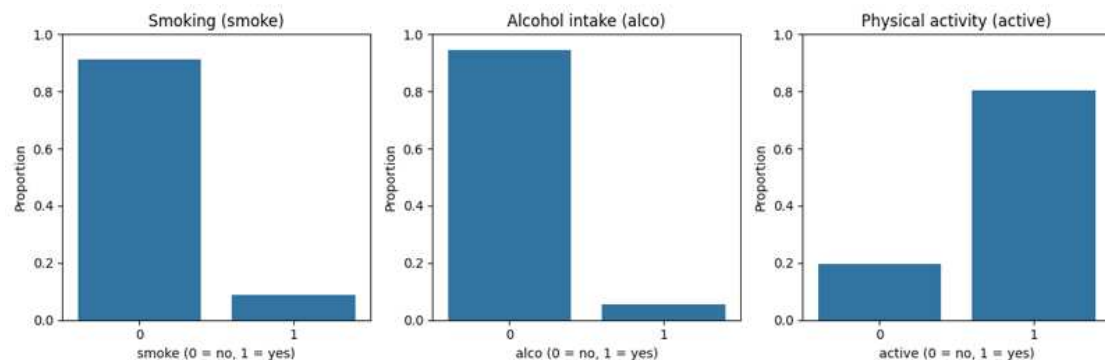


Figure 9: Distribution of lifestyle variables: smoking, alcohol use, and physical activity.

3.5.1 Age and cardiovascular disease

The proportion of disease rises clearly and monotonically with age. In the 20–30 and 30–40 age groups, the disease share is comparatively low. It increases steadily in older groups, with a particularly noticeable increase after age 50. In the 60–70 group, more than half of the records are labelled with cardiovascular disease. This confirms that age is a central predictor in this dataset.

3.5.2 Gender and cardiovascular disease

When examined separately by gender code, the proportion of healthy and diseased records is surprisingly similar: both groups are close to 50% positive. Although the dataset contains more records coded as female, gender alone does not appear to be a strong predictor of `cardio`.

The combined age-and-gender view gives a more nuanced result. Disease rates rise strongly with age in both groups. At younger ages, the male-coded group tends to have a slightly higher disease share; with increasing age, the difference narrows and may partly reverse. The effect of gender is therefore better interpreted in interaction with age than in isolation.

3.5.3 Cholesterol, glucose, and lifestyle factors

Higher cholesterol categories are associated with a higher proportion of positive `cardio` labels. Category 1 has the lowest observed disease share, category 2 is higher, and category 3 is highest. Glucose shows a similar but somewhat weaker pattern: higher categories coincide with higher observed disease prevalence.

Lifestyle variables show weaker marginal relationships with the target than age and clinical variables. The difference between smokers and non-smokers is small in this dataset, as is

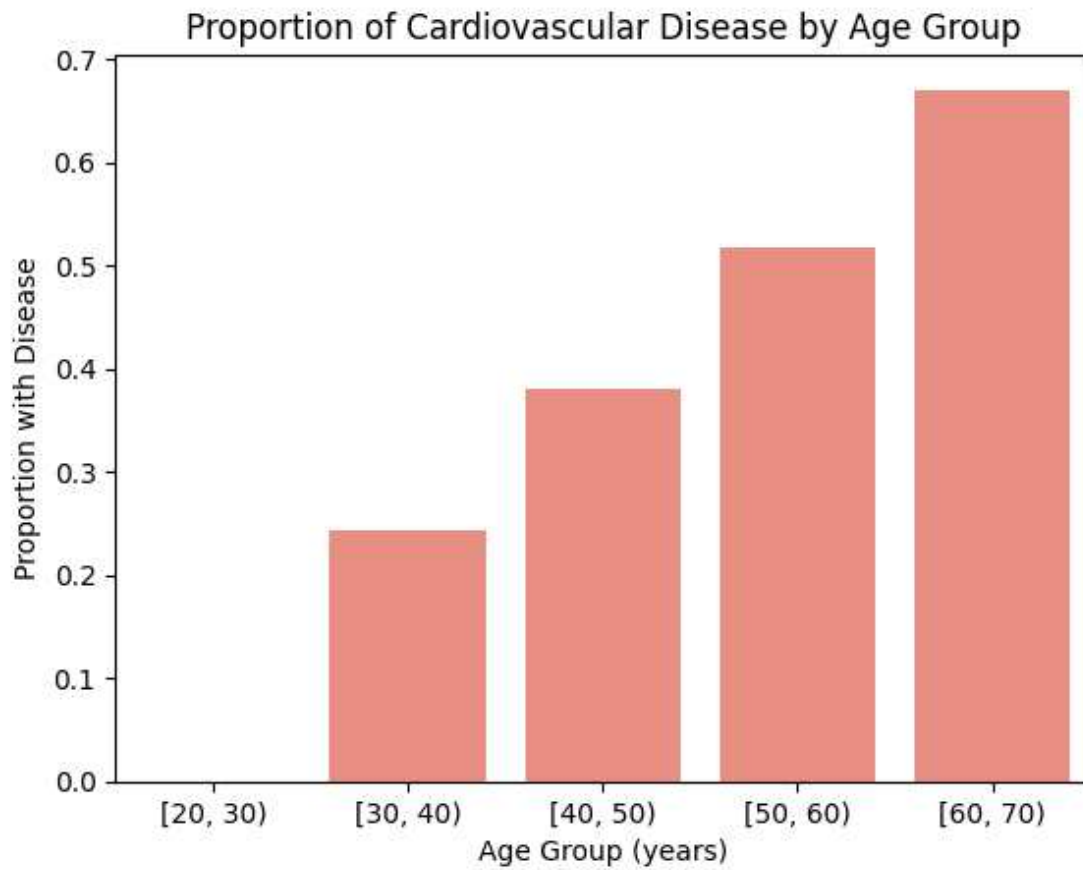


Figure 10: Proportion of cardiovascular disease by age group.

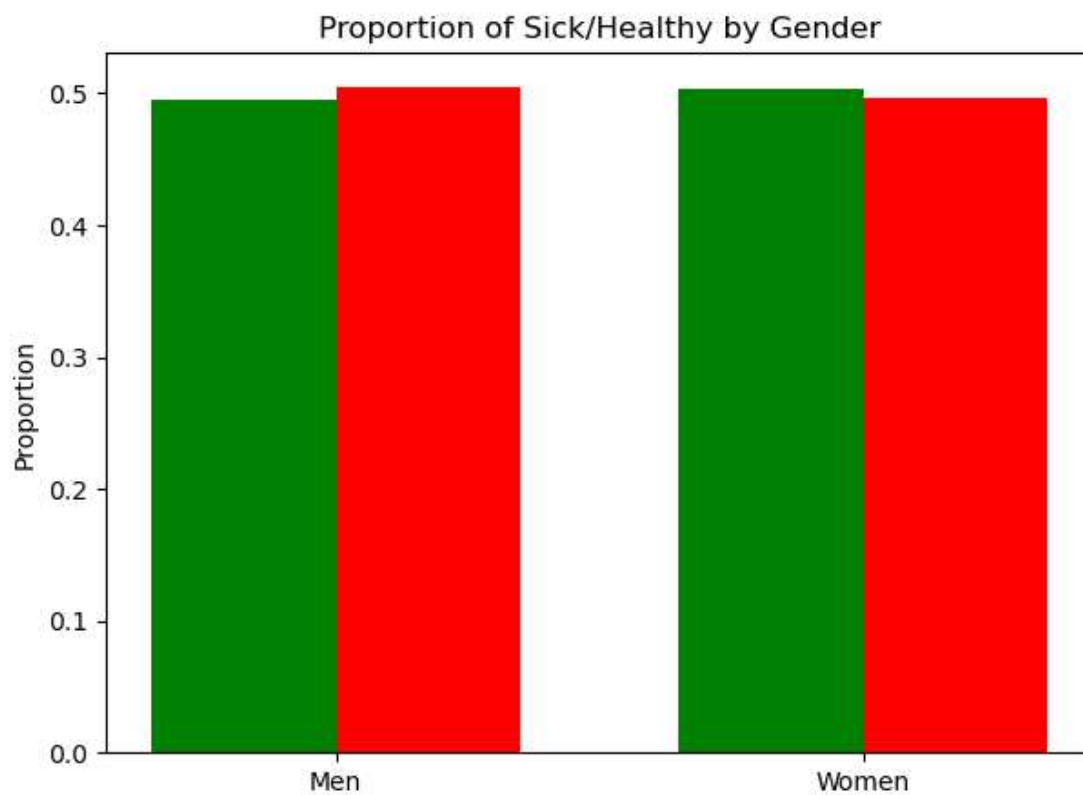


Figure 11: Proportion of healthy and diseased records by gender code.

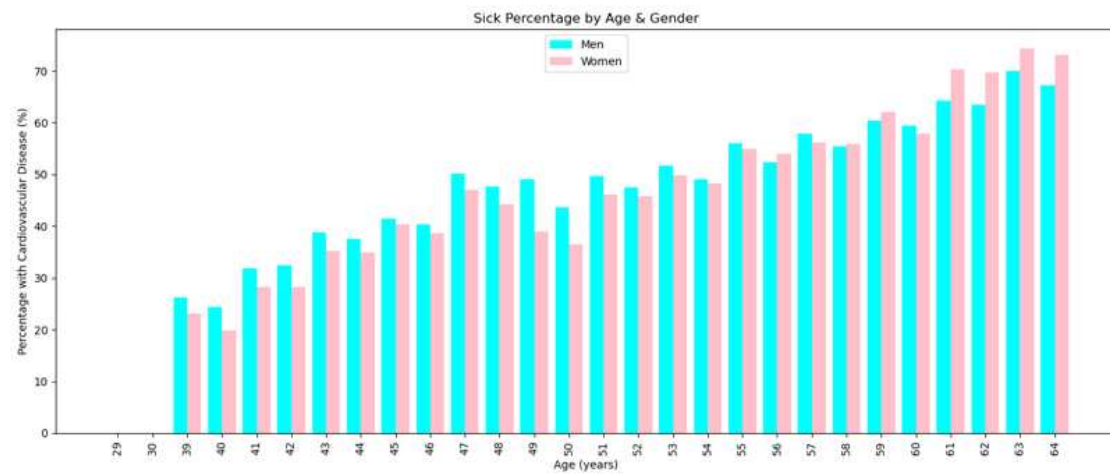


Figure 12: Percentage of cardiovascular disease by age and gender code.

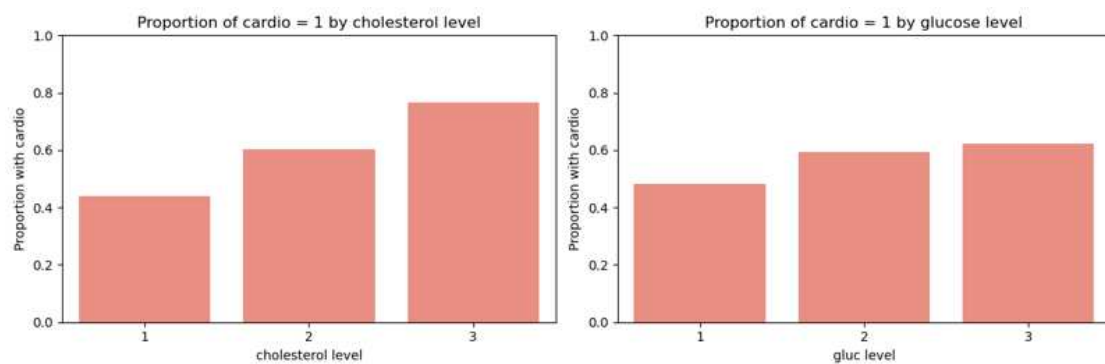


Figure 13: Proportion of cardiovascular disease by cholesterol and glucose category.

the difference between alcohol use and no alcohol use. Physical activity has a small protective association: active records show a slightly lower disease share. These patterns may be confounded, non-linear, or mediated through other variables such as age, BMI, or clinical measurements.

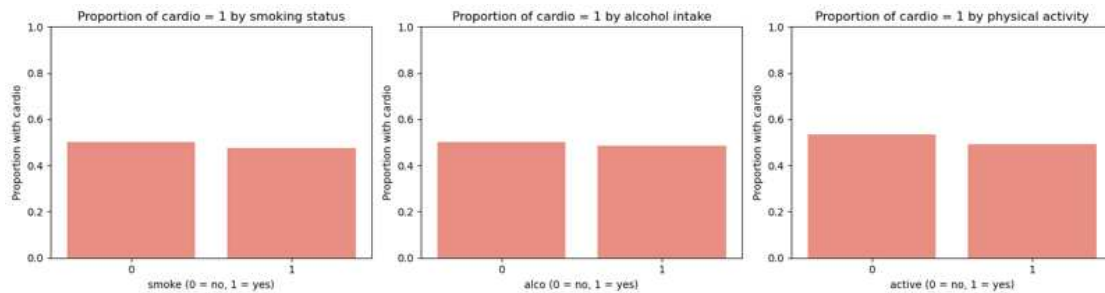


Figure 14: Proportion of cardiovascular disease by lifestyle factor.

3.6 Outlier analysis

The outlier analysis focuses on height, weight, systolic blood pressure, and diastolic blood pressure. These variables contain unrealistic extreme values that are likely to be measurement or data-entry errors. Such values can substantially distort descriptive statistics, correlations, and model performance.

Most heights are concentrated between 150 and 180 cm, but the raw data contains values below 100 cm and above 200 cm. Most weights are between 50 and 110 kg, but some observations lie below 40 kg or above 150 kg, including extreme values that are unlikely for the adult population represented here. Age is more compactly distributed around 45–60 years but still has observations near the boundaries of the range.

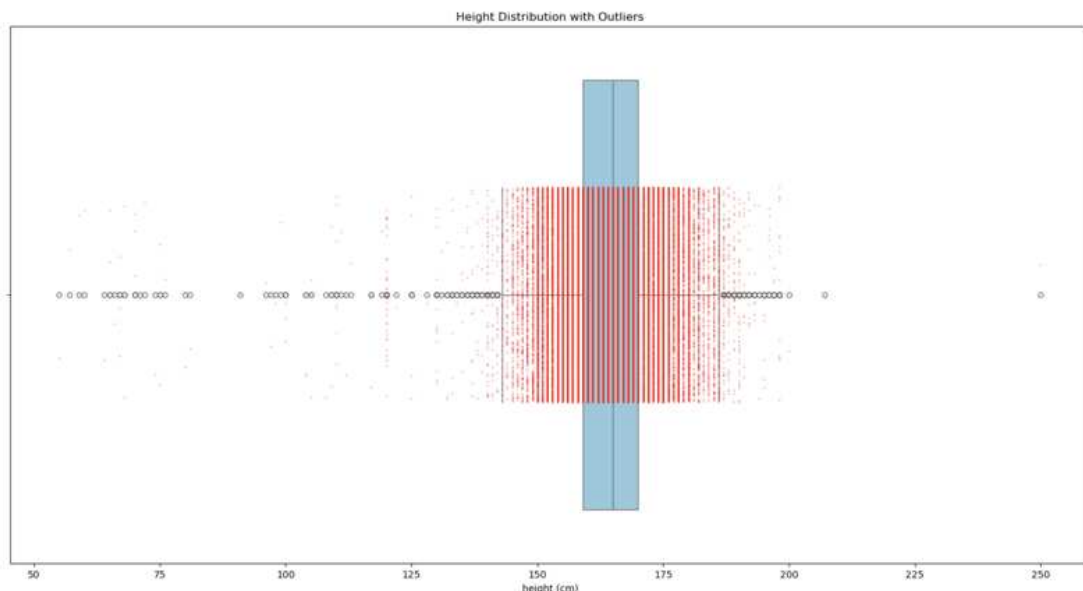


Figure 15: Height distribution with outliers.

Blood-pressure variables show the most serious quality problem. The raw data includes biologically impossible readings, including negative values and values above 10,000 mmHg. It also contains records for which diastolic pressure is greater than systolic pressure. While most values lie in expected physiological ranges, these extreme entries are sufficiently frequent and severe to require explicit cleaning.

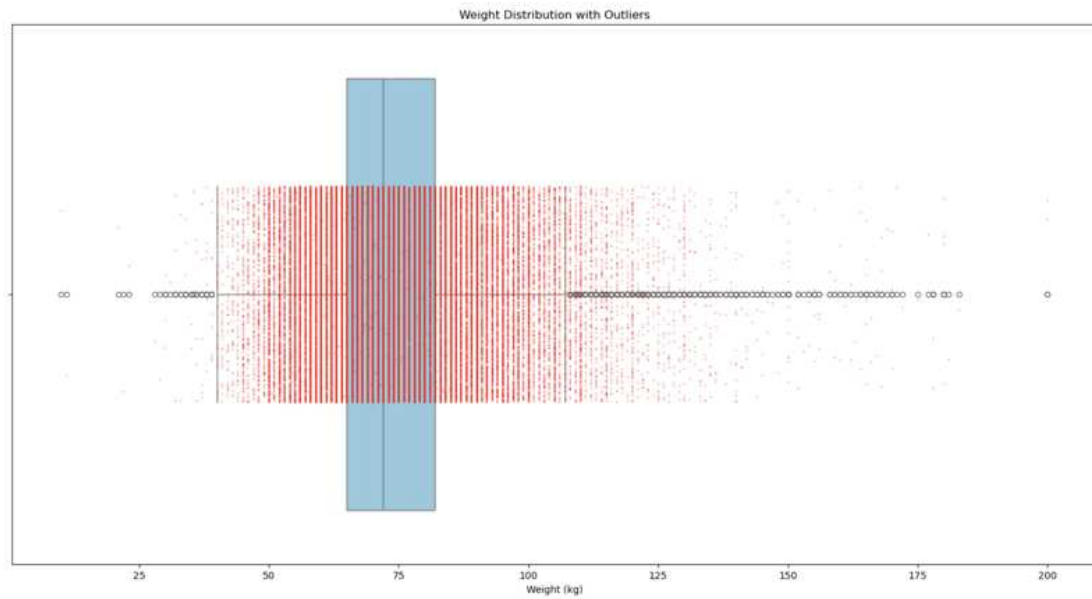


Figure 16: Weight distribution with outliers.

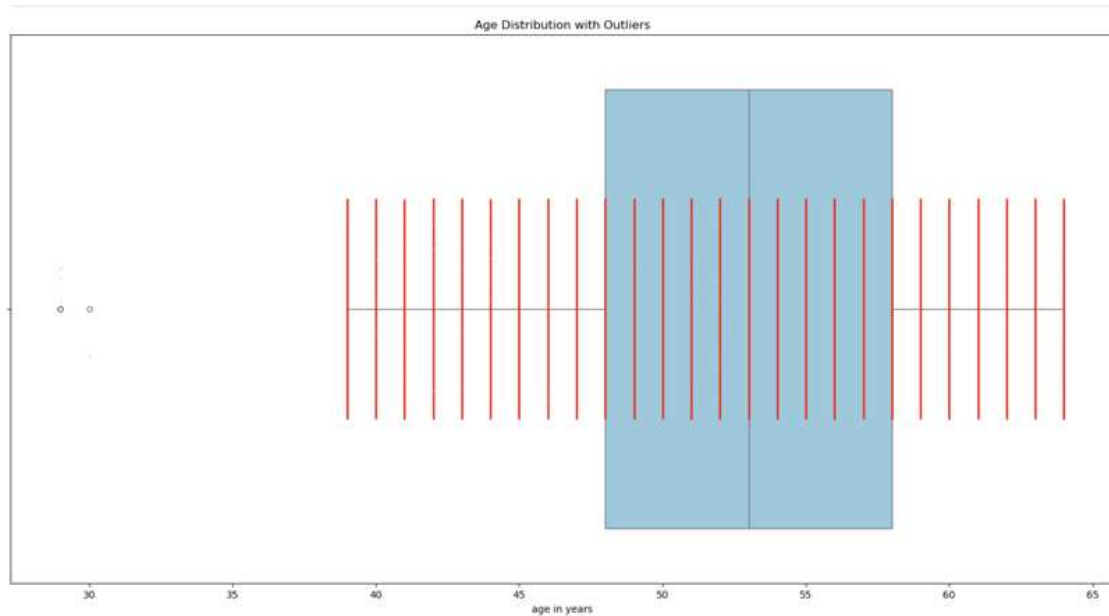


Figure 17: Age distribution in years with outliers.

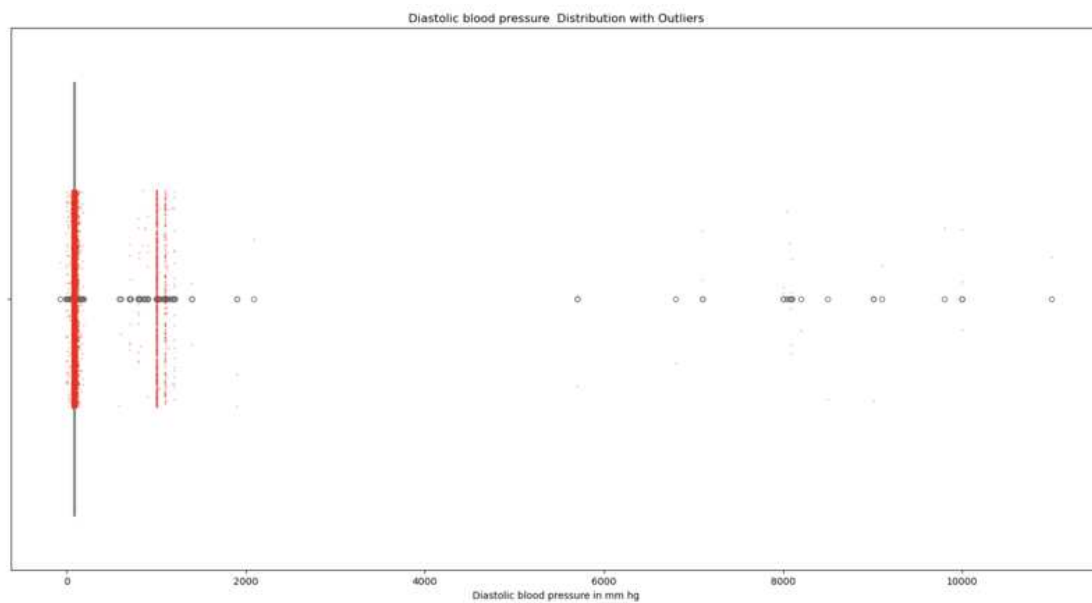


Figure 18: Diastolic blood-pressure distribution with outliers.

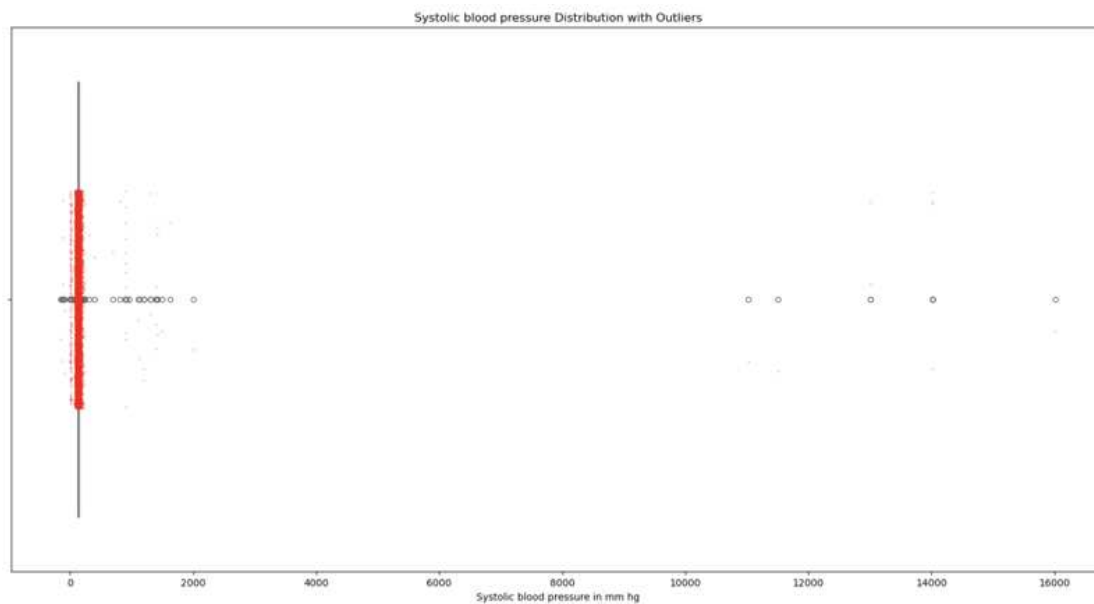


Figure 19: Systolic blood-pressure distribution with outliers.

3.7 Correlation analysis

After the univariate and bivariate analyses, linear relationships between numerical features and `cardio` are examined with correlation analysis. The purpose is to obtain a global view of potential linear associations and the relative importance of individual variables. Correlation does not imply causation.

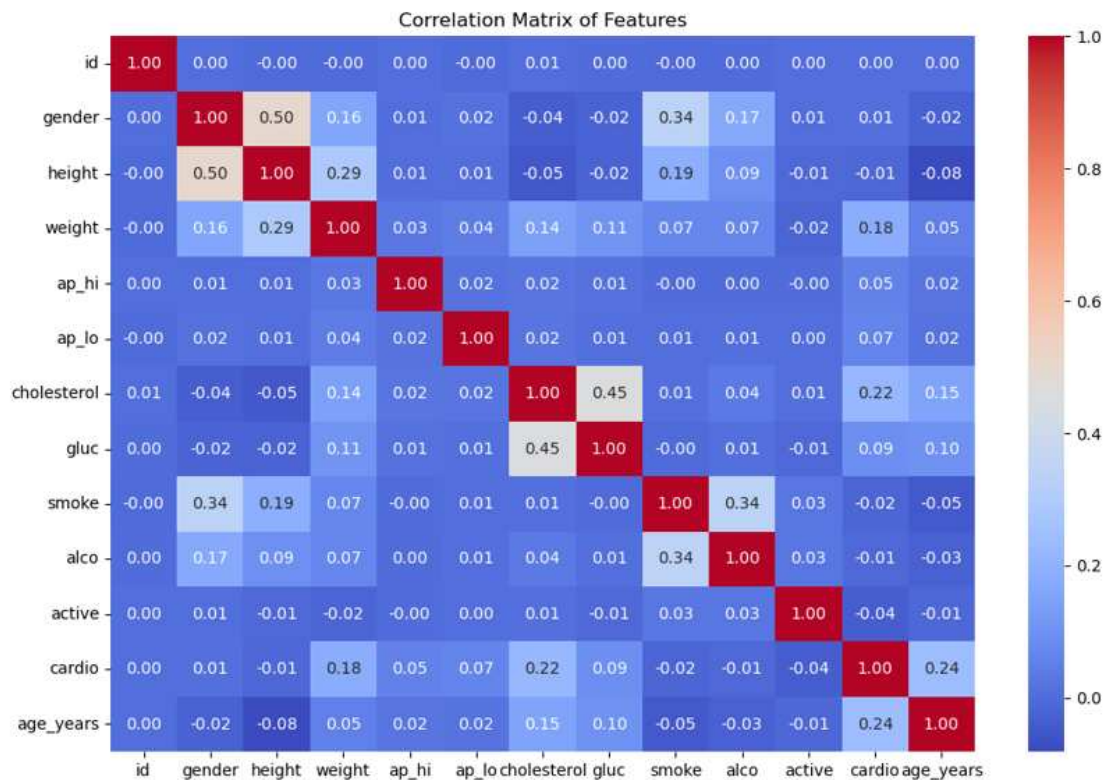


Figure 20: Correlation matrix of numerical features, including `cardio`, before full cleaning.

Most raw correlations are weak to moderate. Age has the strongest positive linear relationship with `cardio` (approximately 0.24), followed by cholesterol (approximately 0.22) and weight (approximately 0.18). Glucose and both blood-pressure variables initially show weak linear correlations. This does not imply that they are medically unimportant; a relationship may be non-linear, hidden by noise, or distorted by outliers.

Gender has almost no linear correlation with `cardio`, consistent with the similar disease proportions observed in the two gender-coded groups. Smoking, alcohol use, and physical activity also have very small correlations in isolation. Their effects may be indirect or may interact with other variables.

3.8 Limitations of the dataset

The dataset has several important limitations:

1. **Data quality and outliers.** No missing values does not mean high quality. The raw data includes implausible heights, weights, and blood-pressure readings, as well as logically inconsistent pressure pairs.
2. **Coded categories rather than continuous tests.** Cholesterol and glucose are three-level categories rather than continuous laboratory values, limiting the available information.

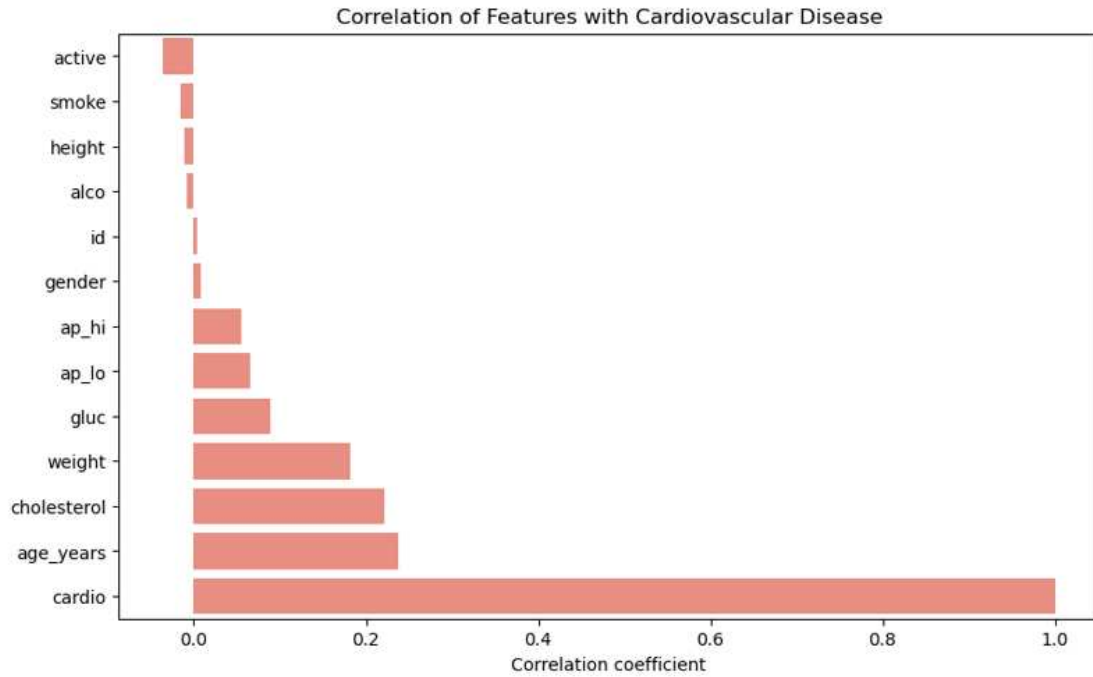


Figure 21: Linear correlation of individual features with `cardio`.

3. **Limited representativeness.** The gender distribution is skewed and most observations are roughly 30–65 years old. The findings are dataset-specific and should not be treated as universal medical rules.
4. **Simplified target.** The binary label cannot represent the heterogeneous diagnoses, severity levels, and stages found in clinical practice.
5. **Correlation rather than causality.** The analysis uses observational data. Statistical associations do not establish cause-and-effect relationships.

4 Data Cleaning and Feature Engineering

Data cleaning is a critical part of the pipeline because the quality of the input data directly limits downstream model performance. The aim is to create a dataset free of clear measurement errors and biologically impossible values while retaining the natural variability of genuine pathological cases.

4.1 Relevant classifications and initial inspection

For validation, the workflow uses established medical concepts. Converting age from continuous days to discrete years supports practical interpretation [5]. BMI is a standard measure of body weight relative to height; extreme values below 15 or above 60 are likely to reflect incorrect height or weight entries [6]. Blood-pressure values must be physiologically possible, and diastolic pressure cannot exceed systolic pressure. Pulse pressure, the difference between systolic and diastolic pressure, is a relevant indicator of arterial stiffness [7].

The raw dataset initially contained 70,000 entries and 13 columns. The initial EDA found that age was stored in days, there were no null values, and data types were mostly consistent integer or floating-point values. It also found major outliers, including heights of 55 cm and negative blood-pressure readings.

4.2 Transformations and new features

The following transformations were applied:

1. **Age conversion.** `age` was divided by 365 and rounded down to create `age_years`. This improves interpretability and comparability with other studies [5].
2. **Removal of the identifier.** The `id` column was removed. As a pure identifier, it has no clinical predictive value and could lead to spurious patterns, overfitting, or data leakage.
3. **Body-mass index.** BMI was computed as

$$\text{BMI} = \frac{\text{weight (kg)}}{(\text{height (m)})^2}.$$

4. **Pulse pressure.** Pulse pressure was computed as

$$\text{PP} = \text{ap_hi} - \text{ap_lo}.$$

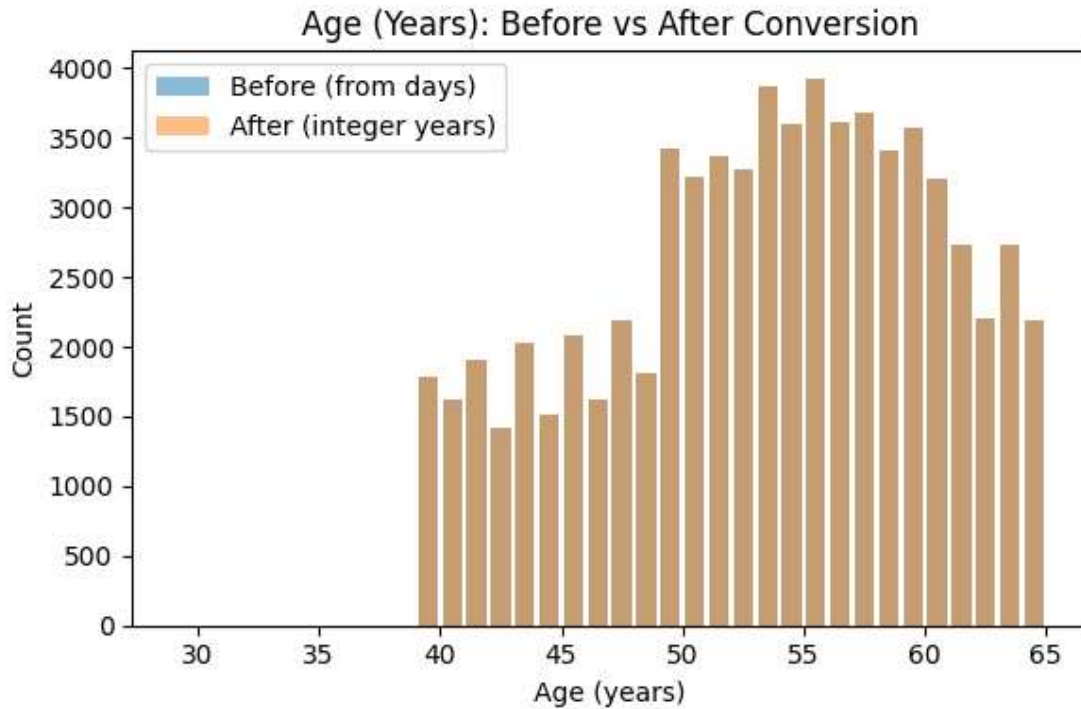


Figure 22: Age distribution before and after conversion from days to years.

4.3 Removal of implausible observations

Conservative thresholds were selected to avoid incorrectly removing plausible severe cases:

- Height: values below 140 cm and above 210 cm were removed.
- Weight: values below 30 kg and above 200 kg were removed.
- BMI: after cleaning height and weight, values below 15 and above 60 were removed.
- Systolic pressure: values outside 80–250 mmHg were removed.

- Diastolic pressure: values outside 40–200 mmHg were removed.
- Logical consistency: records with `ap_lo > ap_hi` were removed.

The blood-pressure plausibility checks are also consistent with processing rules used in national health-survey documentation, which requires a systolic value lower than 300 mmHg and greater than the diastolic value [8].

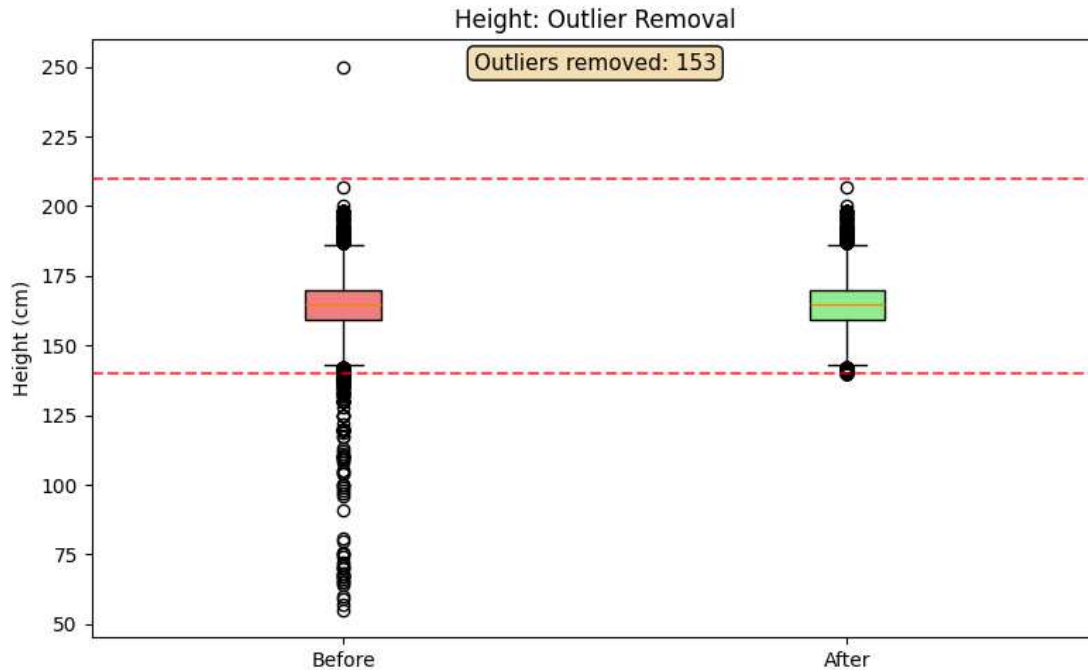


Figure 23: Height outlier removal.

4.4 Outcome of cleaning

The combined cleaning and feature-engineering procedure removed approximately 3–4% of the original dataset. The raw data contained systematic noise in the form of negative blood pressure, values above 10,000 mmHg, and logically inconsistent pressure pairs. No invalid values were found in the categorical variables for gender, cholesterol, glucose, smoking, alcohol use, or physical activity.

After removing extreme outliers, the association between blood pressure and `cardio` becomes considerably clearer. The reported correlation rises from roughly 0.05 to 0.43 for systolic pressure and to 0.34 for diastolic pressure. BMI and pulse pressure add clinically motivated information that was not present as explicit columns in the original dataset.

5 Modelling

5.1 Choice of model

The target variable `cardio` is binary, so classification methods are appropriate. Logistic regression is used as an interpretable baseline. Linear regression would generate continuous predictions that may fall outside the valid probability range of 0 to 1, and it is not designed for binary classification. Logistic regression, by contrast, estimates a probability in the valid range and is therefore suitable for this task.

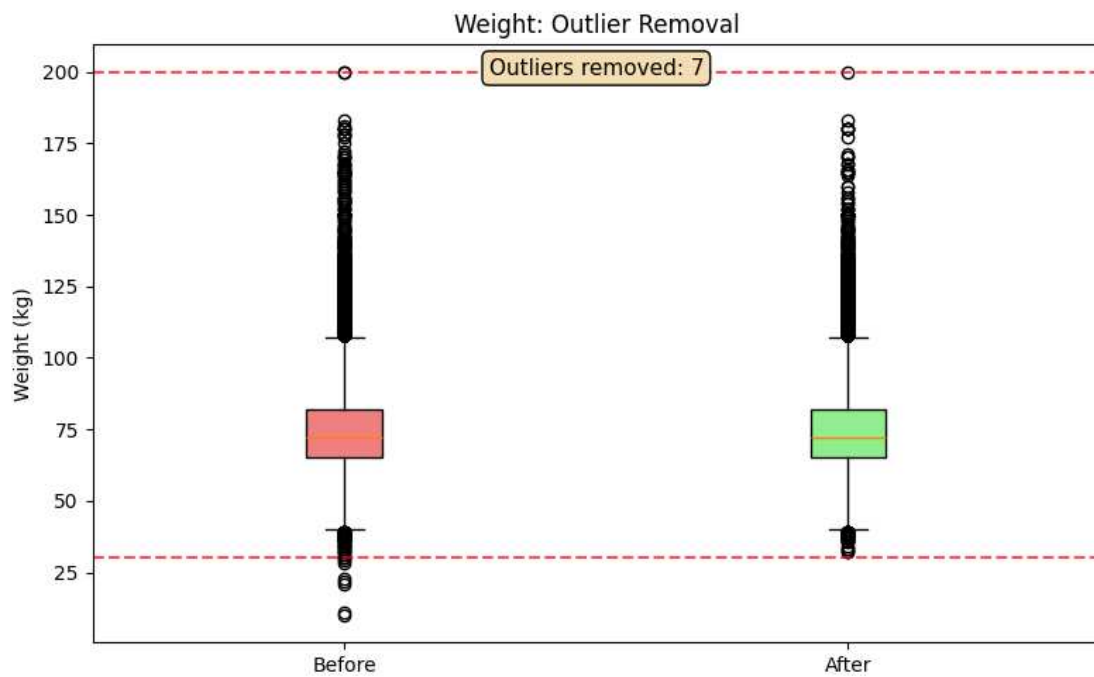


Figure 24: Weight outlier removal.

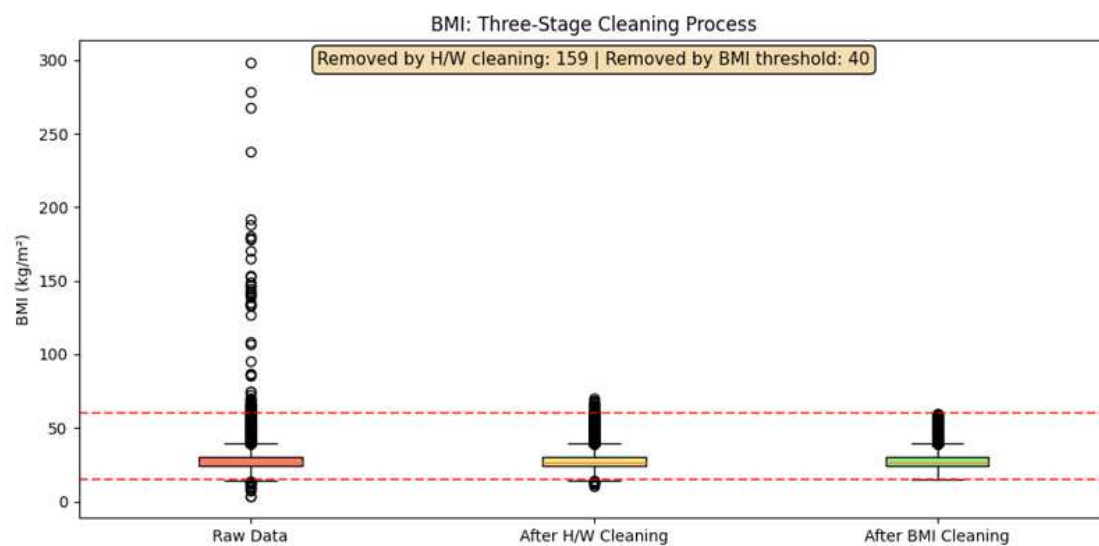


Figure 25: Three-stage BMI cleaning process.

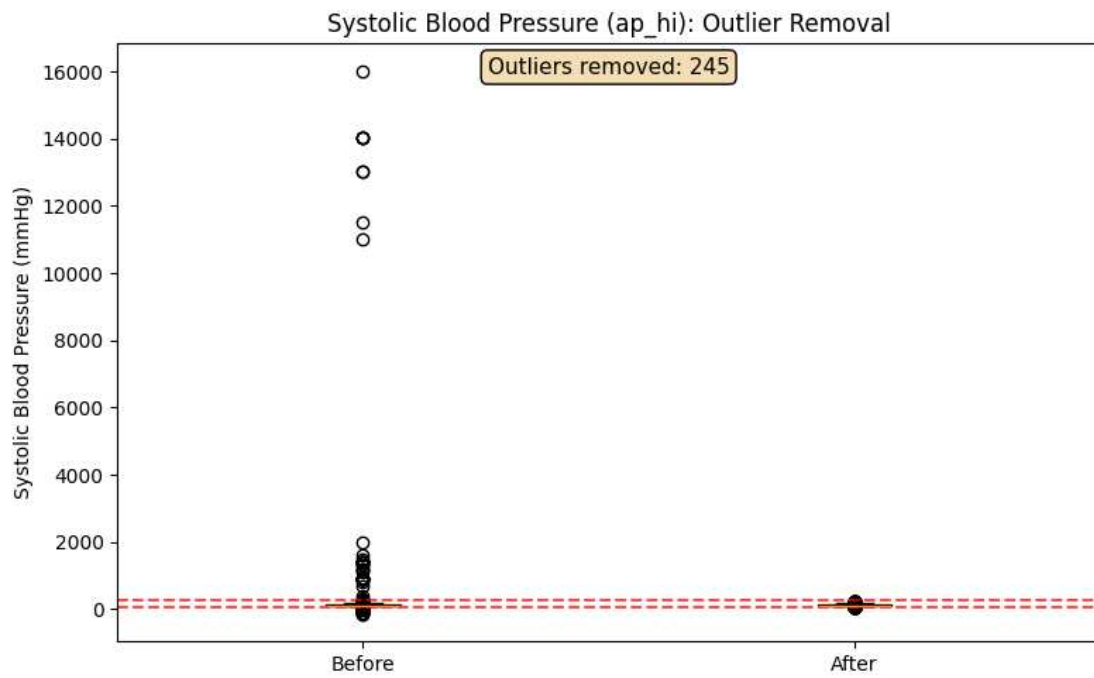


Figure 26: Systolic blood-pressure outlier removal.

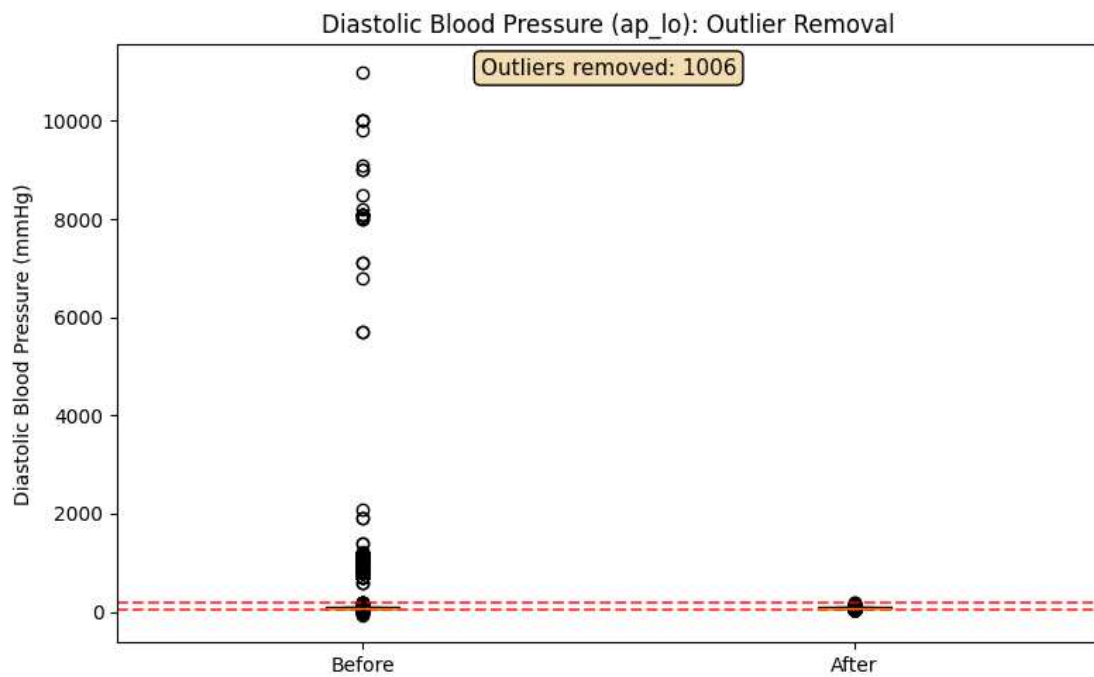


Figure 27: Diastolic blood-pressure outlier removal.

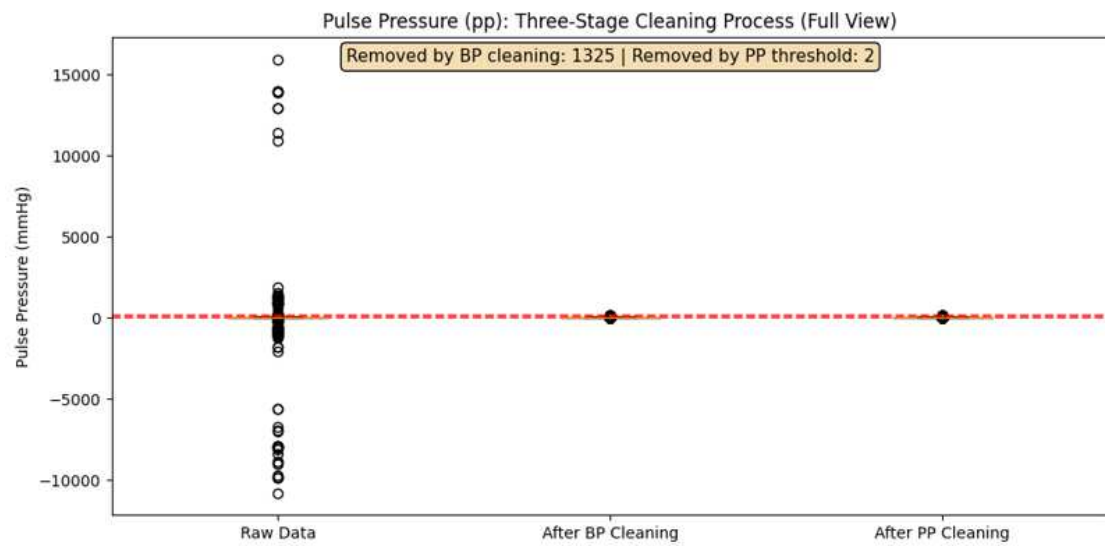


Figure 28: Three-stage pulse-pressure cleaning process, full view.

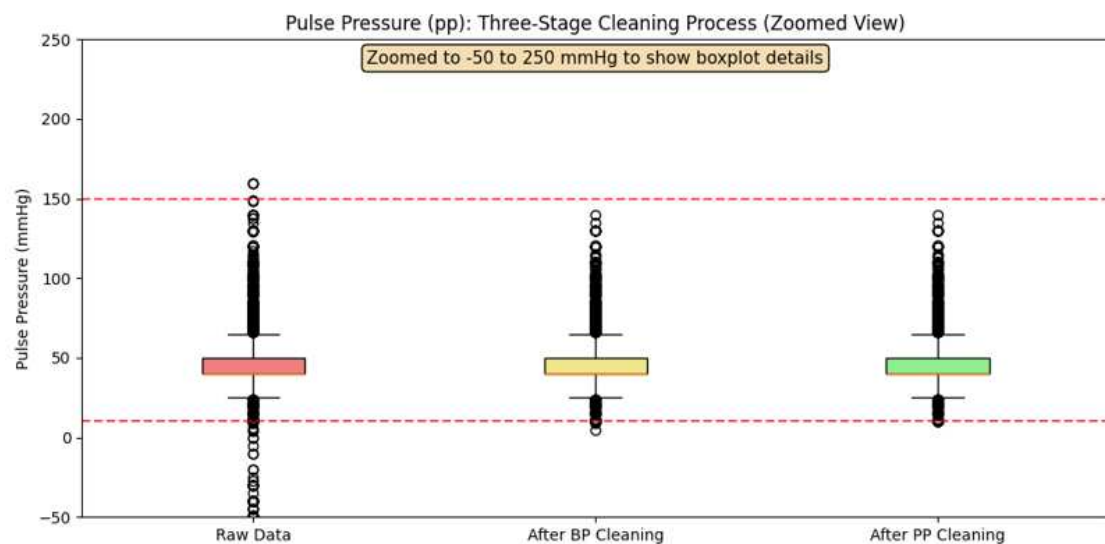


Figure 29: Three-stage pulse-pressure cleaning process, zoomed view.

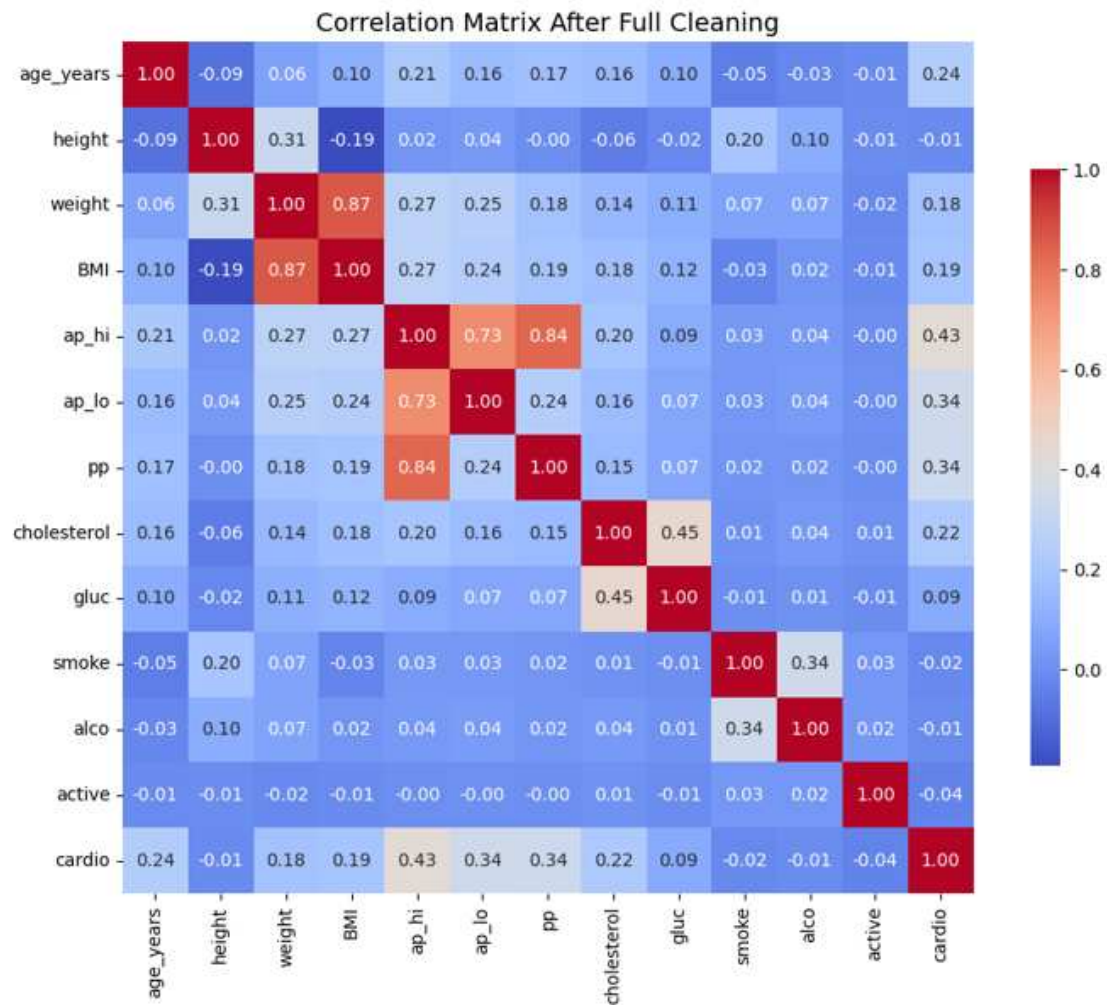


Figure 30: Correlation matrix before and after full cleaning, including BMI and pulse pressure.

5.2 Training procedure

All variables other than the target were used as input features. These include demographic variables, height, weight, BMI, blood pressure, laboratory categories, and lifestyle factors. Since the data was already numerically encoded, no additional categorical encoding was required.

The data was split into 80% training data and 20% test data using `random_state=42`. Stratification was used so that the distribution of `cardio` was the same in training and test sets. This is particularly important in health-related datasets, where unequal class distributions can otherwise lead to biased evaluation.

After the split, the features were standardised to a mean of zero and a standard deviation of one. The scaler was fitted only on the training data and then applied to the test data to avoid data leakage. Logistic regression was trained with `max_iter=1000`, L2 regularisation, the `lbfgs` solver, and `random_state=42`. The same procedure was applied to raw and cleaned variants so that their evaluation remains comparable.

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Train shape:", X_train.shape)
print("Test shape:", X_test.shape)
print("Train target distribution:\n", y_train.value_counts(normalize=True))
print("Test target distribution:\n", y_test.value_counts(normalize=True))
```

Train shape: (54849, 13)
Test shape: (13713, 13)
Train target distribution:
cardio
0 0.504895
1 0.495105
Name: proportion, dtype: float64
Test target distribution:
cardio
0 0.504922
1 0.495078
Name: proportion, dtype: float64

Figure 31: Stratified split of the dataset into training and test sets.

6 Evaluation and Interpretation

6.1 Evaluation metrics

The trained model was assessed with established classification metrics. Accuracy measures the overall share of correct predictions. Precision, recall, and F1 score provide class-specific detail. Recall is especially important here because it measures how many records labelled with disease were identified correctly. ROC-AUC measures the model's ability to distinguish the two classes across classification thresholds.

The metrics are reported for the unfiltered dataset and for the cleaned dataset using the same test parameters. They are not evidence that the model is safe, calibrated, or clinically valid.

6.2 Results with the unfiltered dataset

The unfiltered model achieved overall accuracy of approximately 70%. The detailed metrics show that the negative class (`cardio=0`) was classified somewhat better than the positive class (`cardio=1`). For the negative class, precision was approximately 69% and recall approximately 74%. For the positive class, precision was approximately 72%, but recall was only approximately 67%.

The positive-class recall means that a substantial share of records labelled with disease were classified as healthy. In a real medical setting, this false-negative rate would be serious. The F1 score was approximately 71% for the negative class and 69% for the positive class. ROC-AUC was approximately 0.76, indicating moderate discrimination. Extreme heights, implausible blood pressure, and noisy distributions plausibly make meaningful patterns harder for the model to detect.

Classification Report:				
	precision	recall	f1-score	support
0	0.69	0.74	0.72	7004
1	0.72	0.67	0.69	6996
accuracy			0.70	14000
macro avg	0.71	0.70	0.70	14000
weighted avg	0.71	0.70	0.70	14000

ROC-AUC Score: 0.7629265450372392				
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Figure 32: Logistic-regression evaluation results for the unfiltered dataset.

6.3 Results with the cleaned dataset

After cleaning, the model produced more stable and interpretable results. Removing unrealistic heights, weights, and blood-pressure values created a clearer and more physiologically plausible feature space. Accuracy increased modestly to approximately 72.7%, but the more important change is in the quality of the other metrics and the discrimination score.

Precision for both classes remained similar to the raw-data model. Recall for the negative class increased to approximately 78%, while positive-class recall remained approximately 67%. F1 scores increased slightly to about 74% for the negative class and 71% for the positive class. ROC-AUC increased to approximately 0.80, indicating improved separation between the two classes.

The confusion matrix also shows fewer visibly distorted classifications. The comparison demonstrates that data cleaning can improve a model's ability to detect statistical structure rather than learn from obvious errors and extreme noise. The positive-class recall remains inadequate for any clinical use.

Accuracy: 0.7266827098373806
 ROC AUC: 0.795001018570922

Classification report:

	precision	recall	f1-score	support
0	0.71	0.78	0.74	6924
1	0.75	0.67	0.71	6789
accuracy			0.73	13713
macro avg	0.73	0.73	0.73	13713
weighted avg	0.73	0.73	0.73	13713

Figure 33: Logistic-regression evaluation results for the cleaned dataset.

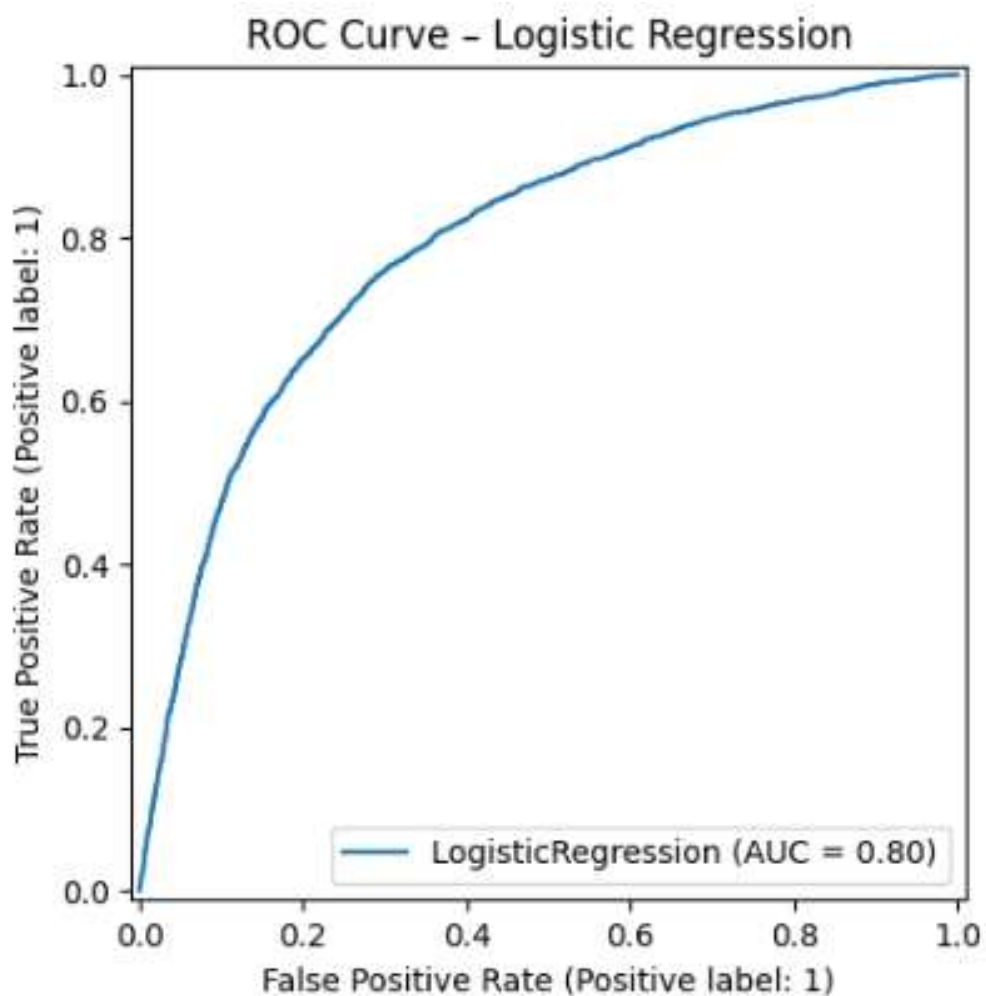


Figure 34: Increase in ROC-AUC to approximately 0.80 after cleaning.

7 Summary, Limitations, and Outlook

7.1 Key findings

The project followed a systematic path from EDA through cleaning and feature engineering to logistic-regression modelling. Its central finding is that data quality materially affects model performance. The original data included biologically impossible values such as negative blood pressure, implausible heights, and logically inconsistent blood-pressure pairs. Cleaning these observations and adding BMI and pulse pressure improved the quality of the analytical basis.

Correlations changed substantially after cleaning. In particular, systolic and diastolic blood pressure, pulse pressure, age, cholesterol, and BMI became more clearly associated with the binary target. The rise in the systolic-pressure correlation from approximately 0.02–0.05 to approximately 0.43 illustrates how severe errors can mask relationships in raw data.

The removal of approximately 3–4% of records with obvious errors led to modestly higher accuracy and a meaningful ROC-AUC increase. These results support the practical lesson that cleaning is not an optional cosmetic step: it is a prerequisite for interpretable modelling on noisy health-related data.

7.2 Limitations

The work has important limitations. It relies on a single, incompletely documented dataset and has no external validation on independent data. Its clinically most important limitation is the positive-class recall of approximately 67%, corresponding to a false-negative rate of about 33% in the unfiltered analysis. Even after cleaning, the model is not suitable for decisions about individual patients.

The results are also limited by the binary target, the coarse coding of cholesterol and glucose, the skewed gender distribution, and the observational nature of the data. These constraints should be stated prominently in any presentation of the project.

7.3 Possible extensions

Future work could extend the data with ethically and legally appropriate clinical variables, such as family history, medication use, and richer laboratory data. External datasets, including data derived from the Framingham Heart Study, could be used to test robustness and generalisability. Researchers have published Framingham-based findings since 1951, making it a valuable source for methodological comparison [3].

Further modelling work should include external validation, calibration assessment, subgroup and fairness analyses, and threshold selection based on the costs of false positives and false negatives. Alternative models may be compared with the logistic-regression baseline, but any improvement should be assessed with transparent, leakage-free validation rather than accuracy alone.

Data and code availability

The raw Kaggle dataset is not included in this document. A repository accompanying this report should provide a link to the source dataset, reproducible notebooks, an environment specification, and instructions for local download. It should contain only code, documentation, and figures that the project team is entitled to publish.

Articles Published Per Decade Based on Framingham Data

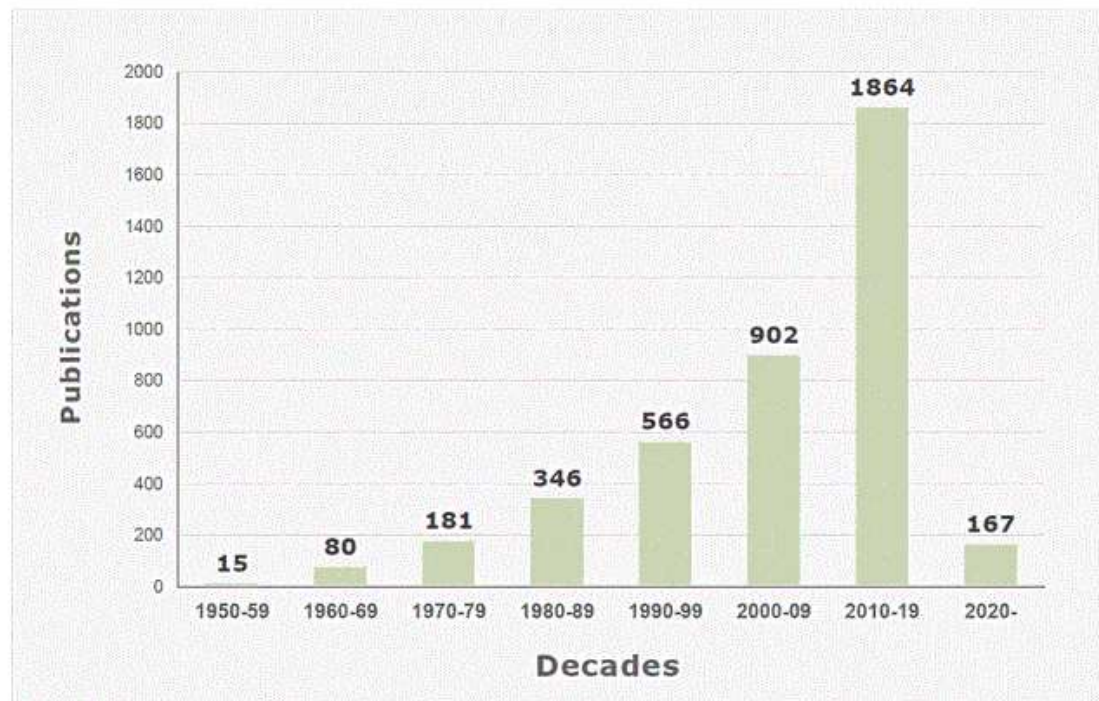


Figure 35: Number of published articles per decade based on Framingham Heart Study data (1950–2020).

References

- [1] S. Ulianova, “Cardiovascular Disease Dataset,” Kaggle. Accessed December 21, 2025. Available: <https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset>
- [2] World Health Organization, “Cardiovascular diseases (CVDs),” July 31, 2025. Available: [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))
- [3] UMM Universitätsmedizin Mannheim, “Framingham Study.” Available: <https://www.umm.uni-heidelberg.de/miism/biomedizinische-informatik/medizinische-statistik-buch/framingham-studie/>
- [4] I. M. Graham et al., “Systematic Coronary Risk Evaluation (SCORE): JACC Focus Seminar 4/8,” *Journal of the American College of Cardiology*, vol. 77, no. 24, pp. 3046–3057, June 2021. doi: <https://doi.org/10.1016/j.jacc.2021.04.052>
- [5] C. Andrade, “Age as a variable: Continuous or categorical?,” *Indian Journal of Psychiatry*, vol. 59, no. 4, pp. 524–525, 2017. doi: https://doi.org/10.4103/psychiatry.IndianJPsychiatry_354_17
- [6] World Health Organization, “Body mass index (BMI).” Available: <https://www.who.int/data/gho/data/themes/topics/topic-details/GHO/body-mass-index>
- [7] T. D. Homan, S. J. Bordes, and E. Cichowski, “Physiology, Pulse Pressure,” *StatPearls*, July 2023.

- [8] Centers for Disease Control and Prevention, “NHANES 2017–March 2020: Blood Pressure—Oscillometric Measurement Data Documentation, Codebook, and Frequencies.” Available: https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2017/DataFiles/P_BPX0.htm